

CPU Codenames Demystified

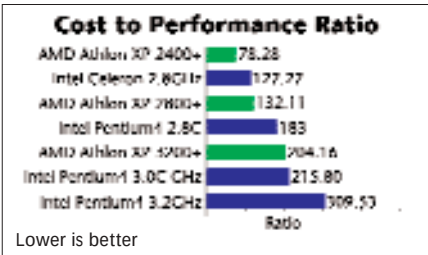
AMD	INTEL
Palomino: The first Athlon XPs from AMD, ranging from the 1500+ to the 2100+, were successors to the aging Athlon Thunderbird series. These processors have an L1 cache of 128 K and an L2 cache of 256 K. The first Athlon XP made its appearance in October 2001.	Tualatin: The last of the Pentium 3 cores. It has an L2 cache of 512 KB. The Celeron 1.4 GHz is a Tualatin with 256 K of L2 cache, and a 100 MHz FSB.
Thoroughbred: The current Athlon XPs available from AMD. These have an L1 cache of 128 K, and an L2 cache of 256 K. This core debuted in June 2002 with the Athlon XP 2200+. It was revised, and the newer Thoroughbred B cores are present on the Athlon XP 2800+, with an FSB of 333 MHz.	Williamette: The original edition of the Pentium 4. It initially had an L1 cache of 128 K and an L2 cache of 256 K. The new Celeron 2.0 GHz is based on this core.
Barton: Thought to be the last in the Athlon XP range of processors. It features the same FSB as the Thoroughbred B, with the L2 cache increased from 256 K to 512 K. The first model starts with the 2800+ tag, and goes on. Currently, available up to 3200+.	Northwood: The Pentium 4 currently available, with 512 K of L2 cache. It's available in two variants—the non-Hyperthreaded 400 MHz FSB version, and the 533 MHz and 800 MHz versions with Hyperthreading. The latest and fastest is the Pentium 4EE, which features the introduction of the 2 MB L3 cache.
Hammer: The latest entrant in the desktop and mobile processing market. It has 512 KB of L2 cache, with a new 940-pin design and 64-bit computing, with native support for existing 32-bit applications. It debuted in October 2003.	Prescott: Originally slated for release in 2003, it's been delayed until 2 February 2004. It's the next fabrication change for the Pentium 4 processors with higher speeds, starting from the Pentium 4 Prescott 2.80A to the Intel Pentium 4 Extreme Edition 3.40 GHz. The L2 cache has been increased to 1 MB for these processors, except for the EE, where the L2 cache has been stepped up to 2 MB.
ClawHammer: This scaled down version of the Hammer will be out by the middle of 2004.	

this case. The Pentium 4 3.2 GHz costs Rs 5,000 more than the Celeron does, and that's for the processor alone!

On the other hand, games and image editing applications such as Adobe Photoshop are resource-hungry; similarly, video encoding applications such as FlaskMPEG and Dr. DivX, or for that matter, Windows Media Encoder, will be more than happy with any resources that the CPU can provide. If these are typical of the applications you intend to run, we recommend at least a Pentium 4 2.8C GHz.

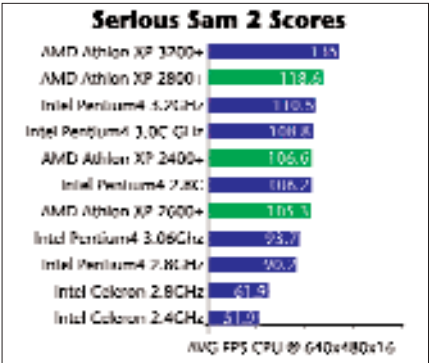
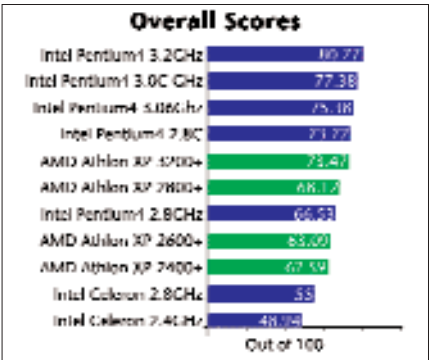
If you'd like the best of both worlds, we suggest an AMD. The Celeron 2.4 GHz costs Rs 6,200. The Athlon XP 2400+ costs Rs 4,900 and gives you a performance gain of approximately 22 per cent. Thus, the 2400+ is not only cheaper but also gives you a better performance.

The blazing performance of AMD 3200+



equalled that of the 2.8C GHz, but fell behind the Pentium 4 3 GHz and 3.2 GHz processors. The Athlon XP 2800+, which is clocked lower than the Pentium 4 2.8 GHz scored better than it, but fell behind the Pentium 4 2.8C GHz and beyond.

The score percentage difference between the AMD 3200+ and the Pentium 4 3.2 GHz is 9 per cent, but the AMD costs a whopping Rs 10,000 less. The architecture and its implementation are at play



AMD Processor Naming Conventions

AMD insists that one should look at performance rather than clock speeds. As a result, clock speeds aren't mentioned; they are indicated, in a relative way, by the number in the processor's name, which is the Performance Rating Number (PRN).

To complicate matters, the Athlon XP PRNs are based on how fast the older Thunderbird Athlon would have to run, to give the same performance of the Athlon XP processor in question. For example, an Athlon XP 1500+ actually runs at 1.33 GHz.

This implies that the Athlon XP 1500+ (1.33 GHz) performs as well as a Thunderbird Athlon running at 1.50 GHz.

An increase of 66 MHz in clock speed translates to an increase of 100 MHz in PRN. Now, this works at lower speeds, but the higher you go, the larger the gap becomes. To address this, AMD has revised its XP rating system a bit. Using the old system, the 2400+ should be running at 1.93 GHz, but actually runs at 2 GHz, reflecting the new rating system.

here—AMD's processors give excellent performance even when clocked at lower speeds. But it's apparent from the scores that the king of the hill is the Pentium 4 3.2 GHz. It retails for Rs 25,000. This is a significant investment for just one part of the computer, but the performance is worth every penny.

If price is no object, check out the face-off between the Pentium 4 EE and the AthlonFX.